

Customer Retention & Churn Analysis

Future Interns – Data Science & Analytics Task 2

1. Objective

The objective of this project is to analyze customer retention and churn behavior using customer subscription data. The analysis aims to identify factors that influence customer churn, understand customer lifetime patterns, and provide business recommendations to improve retention.

2. Dataset Overview

The analysis was performed using the Telco Customer Churn Dataset. The dataset contains customer demographic information, subscription details, service usage, billing information, and churn status.

Key attributes include:

- Customer ID
- Gender
- Senior Citizen Status
- Tenure analysis
- Contract Type
- Internet Service
- Monthly Charges
- Total Charges
- Payment Method
- Churn Status

3. Data Preparation

The dataset was imported into Tableau for analysis.

The following preparation steps were performed:

- Verified data types of all columns
- Reviewed customer records
- Organized dimensions and measures
- Created calculated fields for:
 - Churn Rate
 - Customer Lifetime Value (CLV)
- Created tenure bins for retention analysis

4. Dashboard Analysis

The dashboard was designed to monitor customer retention and identify churn trends.

The dashboard includes:

- Total Customers KPI
- Churn Customers KPI
- Churn Rate KPI
- Contract Type vs Churn
- Internet Service vs Churn
- Tenure Analysis
- Monthly Charges Analysis
- Customer Lifetime Value Analysis

5. Key Findings

5.1 Contract Type Analysis

Customers with Month-to-Month contracts showed the highest churn rate. Customers with longer contract commitments demonstrated better retention.

5.2 Internet Service Analysis

Certain internet service categories experienced higher churn compared to others, indicating possible differences in customer satisfaction and service value.

5.3 Tenure Analysis

Customers with shorter tenure periods were more likely to churn. Customers who remained with the company for longer periods showed significantly lower churn rates.

5.4 Monthly Charges Analysis

Customers with higher monthly charges displayed a greater tendency to leave the service compared to customers with moderate charges.

5.5 Customer Lifetime Value (CLV)

Retained customers generated substantially higher Customer Lifetime Value compared to churned customers. This highlights the financial importance of customer retention.

6. Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Encourage customers to move from Month-to-Month plans to annual or long-term contracts through discounts and promotional offers.
2. Implement targeted retention campaigns during the first year of customer onboarding.
3. Introduce loyalty and rewards programs for long-term customers.

4. Monitor customers with high monthly charges and provide personalized offers to reduce churn risk.
5. Improve customer engagement and service quality to increase retention and customer satisfaction.

7. Conclusion

This project successfully analysed customer retention and churn patterns using Tableau. The findings indicate that contract type, tenure, and monthly charges are significant factors influencing customer churn.

By focusing on customer retention strategies, businesses can reduce churn, increase customer lifetime value, and improve long-term profitability. The dashboard provides valuable insights that can support data-driven decision-making and customer retention initiatives.